Lab 01 | Event Analysis

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The analysis of this packet capture (PCAP) file took a while and used a variety of tools, filters, and settings. Wireshark was the most prominent tool used for this task, as it provided a nearly complete view of all the addresses, connections, protocols, and packet data for me to base my investigation and conclusion around. To begin, we were presented with the raw packet data and the list of things to check for in our analysis in the instructions. Naturally, I began from the top with letter a) – the length of the packet capture session and other general data. I was able to find this and a few other statistics quickly by heading to Statistics > Capture File Properties (1). I just had to scroll down, and I was able to see the dates of the first and last packets recorded, number of packets recorded in total, and number of bytes recorded in total, all of which correspond to the first few items in our checklist. Next, I went to identify which protocols were being used at the time of the session. This was also a simple task; this information was discoverable by going to Statistics > Protocol Hierarchy. Upon reading through the protocols, I found that the ones used were the Internet Protocol v4 (IPv4), User Datagram Protocol (UDP), Transmission Control Protocol (TCP), Hypertext Transfer Protocol (HTTP), Transport Layer Security (TLS), File Transfer Protocol (FTP), and the Address Resolution Protocol (ARP) (2). Next is one of the most important questions and that is: when was the majority of the traffic recorded? There were two points that showed high levels of recorded traffic in the capture, that being the period between 17:30:00-17:32:30 and the small period between 17:37:35-17:40:00. I discovered this by looking at a graph of the entire capture. I found this graph by going to Statistics > I/O Graphs (3). This graph shows the entire timeline of the capture and denotes two major spikes in packet traffic around the aforementioned time periods. These next two graphs highlight the specific times up close after zooming in (4)(5). Note that I did not edit any of the tick box filters as I did not find it necessary.

Now that we have established some basic information about the packet capture, we can investigate the more technical part of this section. Letter f) asks what caused this transmission spike, and I have a worrying guess as to what that might be. What caused most of the traffic to occur in the first period was trending towards what looks like unauthorized access, Media Access Control (MAC) spoofing, and ARP poisoning. I will address this later, but the hard facts are as follows: From the 17:31:00-17:32:00 mark, we find the host accessing sites and services that are not merely checking email. From packets 1526-1571, we find the usage of FTP and successful file transfers, along with multiple logins to a Linux wireless local area network (WLAN) server on the same device. This alone suggests that someone has gained unauthorized access to the client's host (6). Moreover, this is proven by plugging the PCAP file into NetworkMiner, a separate tool from Wireshark, that gives us details on a variety of different metrics on the PCAP, but in this point, specifically hosts. The IPv4 address "66.39.22.157" is the address of the aforementioned Linux WLAN server; it is possible that this server is part of the home network of the client, but it is far more likely that this is an external mail server being accessed via the Internet Service Provider (ISP) connection. Both of those pieces of information were found by

just clicking on the Hosts tab, then scrolling down to the individual addresses, then finally, by clicking on the drop-down arrow next to the address (7)(8). This was not all, however. There were also a number of ARP broadcasts made to the client's entire home network, with the gateway identifying the devices on the network multiple times over not just this period, but throughout the capture (9). This appears to be enumeration considering the frequency of the requests to what merely appears to the gateway to be the host device.

Continuing down the checklist, I also analyzed the ISP sites that were accessed during the capture. Throughout the capture, I noticed that there was only one such site that the host connected to, and that was America Online (AOL). The IPv4 address for the AOL server is "152.163.15.208" and that information became available after I switched back to Wireshark and went to Statistics > Resolved Addresses. I just typed in "aol" in the search bar, and in the screenshot provided, there are only a fraction of the total number of results that appeared (10). You can see in the screenshot, though, that the server address that was frequently seen in the capture is present in the results. There were also other websites' domain names that were resolved, like Yahoo, Yahoo Mail, MSN, and RBFCU. Moving on, finding the host name and the IP address proved to be another easy task by conferring with NetworkMiner. Since most of the traffic we can see in the capture came from the address "172.16.1.35", I decided to check the Hosts tab in NetworkMiner to see what this was, and voila, it was the client's host computer. Its name is KaufmanUpstairs, and this is visible in the screenshot (11). Also visible in the screenshot is the operating system of this device, which is Windows 2000 Professional.

Onto giving a general overview of the home network. We have a total of four devices to cover, and this information was acquired by checking the Hosts tab in NetworkMiner once again, as well as using a feature in Wireshark. I first went to Statistics > Endpoints to find the network addresses of all devices on the network. I compared this to the packet capture itself to see which of these devices the host accessed by just seeing if there were packets with the source or destination being one of them and the host. There were The first device is the router, or default gateway of the network. Its name is "homeportal.gateway.2wire.net" and its IPv4 address is "172.16.0.1". This screenshot shows the information (12). I ascertained that this is the router because of the fact that the ARP broadcast requests were being relayed through it. Second is the host, or the client's main computer. This device is called KaufmanUpstairs with an IPv4 address of "172.16.1.35". I have already described above my process of recognizing this as the host. The third device is a DVR device named "DVR-8525.local" with an IPv4 address of "172.16.1.37", whose information can be found at the following screenshot: (13). This device was contacted a couple of times throughout the capture, but I did not deem it to be noteworthy as part of the forensics process. It was also one of the devices that was enumerated through the many ARP requests sent over the 8 and a half minutes. The last device I noticed was the device: "KAUFMANUPSTAIRS". This is an IP network camera with an IPv4 address of "176.16.1.39".

This was also identified in the ARP scan and its information can be found in the following screenshot (14). The camera shares a MAC address with the host computer, and the home network connects to the gateway to gain access to the Internet.

Before identifying all the key information I could gather and giving the client my final report, I also took the important step of refreshing my knowledge on Wireshark to make sure I could provide the most accurate and effective information to him. I took the Wireshark Training Module on SimSpace and completed it (15)(16).

Onto the main topic: The situation. This began when a client reached out to me suspecting that a security event had occurred at his home. He mentioned that he only uses his Internet access to read email, and he was running Wireshark when this event occurred. He provided me with the PCAP file and before I began to give my verdict on the potential incident, I needed to do something important first, which was to gather the previously described information and compile it.

The most interesting part of this case begins with the spiking in packets happening at 17:30:00-17:32:30. There was a high number of packets around this time, and it is the period that most drew my attention when beginning to look at the file. What I discovered was that there was a considerable variety of traffic going on that all pointed to the aforementioned conclusions: unauthorized access, MAC spoofing, and ARP poisoning were occurring. I will briefly mention other instance was the other packet spike I came across at 17:37:30-17:38:00, there was another high volume of packets recorded. 1563 packets were recorded during this time period, and it was the host device communicating with an AOL server, whose address was “153.163.15.208”. Since AOL was the ISP for my client, the connection itself is not particularly suspicious, but the packet quantity is what is of greater concern. Thousands of bits of data were being captured in these final thirty seconds of the file, contributing to the spike, but it is unclear what the nature of this data was from the resources I used to analyze this capture.

Onwards to the first item, the unauthorized access. Across the early parts of the capture, we can see multiple logon attempts to the Linux WLAN server, which would be easily explained if the user was just misspelling the password and getting denied access, but it was multiple successful attempts by different users, as can be seen by the 230 response from the server, mentioning that there are simultaneous users logged on. The unauthorized access and usage is characterized by web surfing and file transfers. The first instance we see of it is going to MSN.com shortly after the login of simultaneous user #3/32.

Second, the MAC spoofing. The amount of ARP requests appear to have originated from the host itself, as can be seen in every iteration of ARP request in the file. The entire home network was reached many times throughout the capture, whether that was in a high-traffic period of time like the early and late traffic spikes, or the milder periods in between the peaks.

What leads me to believe there is MAC spoofing is that there would not be a reason for multiple users to be logging in to the server and communicating with the entire home network repeatedly over the eight minutes. This was complimented by the discovery beacon responses given as well, giving the names of the devices it reached. This seems to be a move by the attacker to find as many holes as possible on the network besides the one they have already found: the credentials of the client.

Appendix

1.

Wireshark · Capture File Properties · Lab1PCAP.pcap

Details

First packet:	2005-10-30 17:29:35
Last packet:	2005-10-30 17:38:00
Elapsed:	00:08:25

Capture

Hardware:	Unknown
OS:	Unknown
Application:	Unknown

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit
Unknown	Unknown	Unknown	Ethernet	65535 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	2449	1563 (63.8%)	2 (0.1%)
Time span, s	505.697	146.613	2.260
Average pps	4.8	10.7	0.9
Average packet size, B	331	276	798
Bytes	811157	431186 (53.2%)	1596 (0.2%)
Average bytes/s	1604	2940	706
Average bits/s	12k	23k	5650

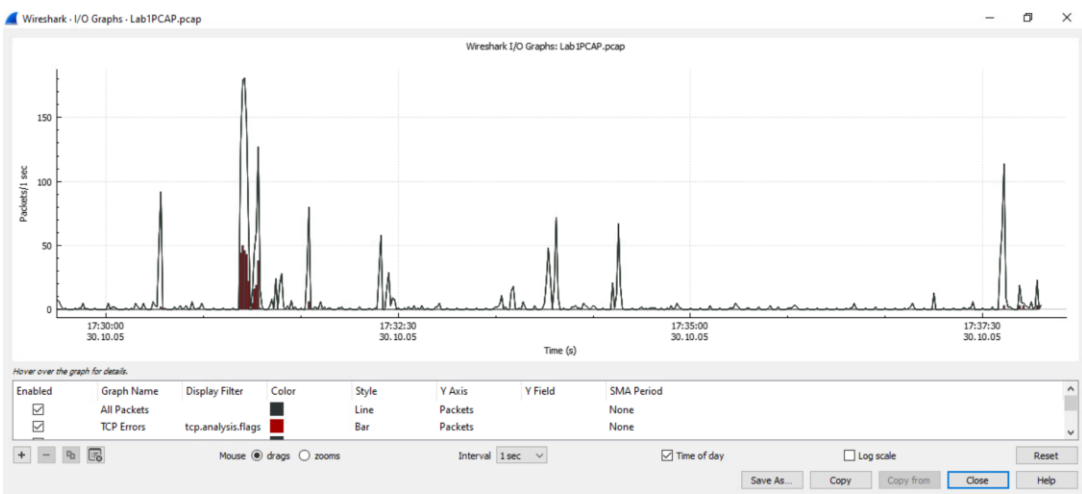
2.

Wireshark · Protocol Hierarchy Statistics · Lab1PCAP.pcap

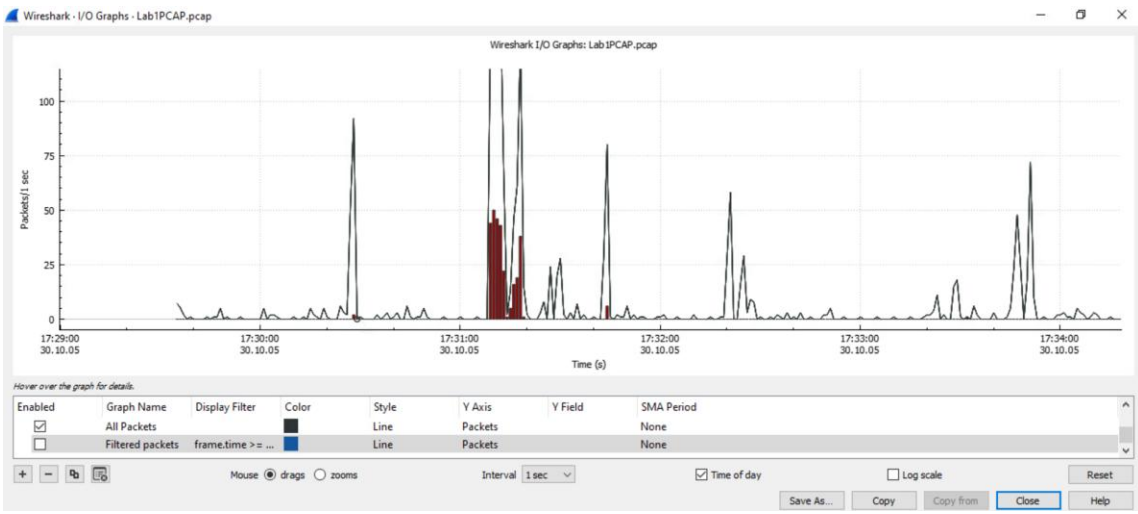
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	1563	100.0	431186	23k	0	0	0
Ethernet	100.0	1563	5.1	21882	1194	0	0	0
Internet Protocol Version 4	97.8	1528	7.1	30560	1667	0	0	0
User Datagram Protocol	6.3	98	0.2	784	42	0	0	0
TiVoConnect Discovery Protocol	2.2	34	1.3	5662	308	34	5662	308
NetBIOS Name Service	0.4	6	0.1	300	16	6	300	16
Multicast Domain Name System	0.1	2	0.1	352	19	2	352	19
Dynamic Host Configuration Protocol	0.4	6	0.4	1800	98	6	1800	98
Domain Name System	3.2	50	1.9	8281	451	50	8281	451
Transmission Control Protocol	91.5	1430	83.0	357805	19k	1166	259330	14k
Transport Layer Security	4.5	71	6.3	27043	1475	71	27043	1475
Hypertext Transfer Protocol	8.6	135	53.1	228816	12k	116	69718	3804
Media Type	0.2	3	20.1	86560	4723	3	87960	4799
Malformed Packet	0.1	1	0.0	0	0	1	0	0
Line-based text data	0.7	11	24.0	103608	5653	11	34530	1884
Compuserve GIF	0.3	4	7.7	33304	1817	4	33511	1828
FTP Data	0.6	10	3.0	13096	714	0	0	0
Line-based text data	0.6	10	3.0	13096	714	10	13096	714
File Transfer Protocol (FTP)	3.1	48	0.4	1720	93	48	0	0
Address Resolution Protocol	2.2	35	0.4	1520	82	35	1520	82

Display filter: frame.time >= "2005-10-30 17:30:00" && frame.time

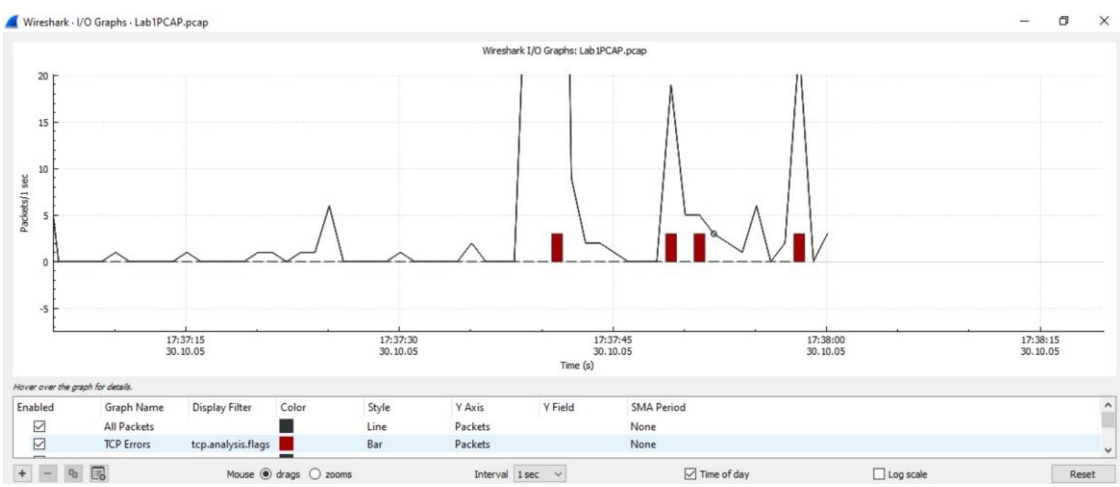
3.



4.


Wireshark · I/O Graphs · Lab1PCAP.pcap
Wireshark I/O Graphs: Lab1PCAP.pcap
Packets/1 sec
100
75
50
25
0
17:29:00 30.10.05
17:30:00 30.10.05
17:31:00 30.10.05
17:32:00 30.10.05
17:33:00 30.10.05
17:34:00 30.10.05
Time (s)
Hover over the graph for details.
Enabled
Graph Name
Display Filter
Color
Style
Y Axis
Y Field
SMA Period
All Packets
Line
Packets
None
Filtered packets
frame.time >= ...
Line
Packets
None
Mouse
drags
zooms
Interval
1 sec
Time of day
Log scale
Reset
Save As...
Copy
Copy from
Close
Help

5.


Wireshark · I/O Graphs · Lab1PCAP.pcap
Wireshark I/O Graphs: Lab1PCAP.pcap
Packets/1 sec
20
15
10
5
0
-5
17:37:15 30.10.05
17:37:30 30.10.05
17:37:45 30.10.05
17:38:00 30.10.05
17:38:15 30.10.05
Time (s)
Hover over the graph for details.
Enabled
Graph Name
Display Filter
Color
Style
Y Axis
Y Field
SMA Period
All Packets
Line
Packets
None
TCP Errors
tcp.analysis.flags
Bar
Packets
None
Mouse
drags
zooms
Interval
1 sec
Time of day
Log scale
Reset

6.

No.	Time	Source	Destination	Protocol	Length	Info
1547	17:32:25.464357	172.16.1.35	66.39.22.157	FTP	62	Request: TYPE A
1548	17:32:25.542997	66.39.22.157	172.16.1.35	FTP	70	Response: 200 Type okay.
1549	17:32:25.549998	172.16.1.35	66.39.22.157	FTP	60	Request: PASV
1550	17:32:25.625830	66.39.22.157	172.16.1.35	FTP	104	Response: 227 Entering Passive Mode (66,39,22,157,238,164)
1551	17:32:25.626436	172.16.1.35	66.39.22.157	TCP	62	3537 → 61092 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM=1
1552	17:32:25.702625	66.39.22.157	172.16.1.35	TCP	60	61092 → 3537 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
1553	17:32:25.702684	172.16.1.35	66.39.22.157	TCP	54	3537 → 61092 [ACK] Seq=1 Ack=1 Win=65535 Len=0
1554	17:32:25.702809	172.16.1.35	66.39.22.157	FTP	60	Request: LIST
1555	17:32:25.781790	66.39.22.157	172.16.1.35	FTP	128	Response: 150 Data connection accepted from 68.92.158.179:3537; transfer starting.
1556	17:32:25.791557	66.39.22.157	172.16.1.35	FTP-DATA	1506	FTP Data: 1452 bytes (PASV) (LIST)
1557	17:32:25.799613	66.39.22.157	172.16.1.35	FTP-DATA	1506	FTP Data: 1452 bytes (PASV) (LIST)
1558	17:32:25.799664	172.16.1.35	66.39.22.157	TCP	54	3537 → 61092 [ACK] Seq=1 Ack=2905 Win=65535 Len=0
1559	17:32:25.886135	66.39.22.157	172.16.1.35	FTP-DATA	1506	FTP Data: 1452 bytes (PASV) (LIST)
1560	17:32:25.893982	66.39.22.157	172.16.1.35	FTP-DATA	1506	FTP Data: 1452 bytes (PASV) (LIST)
1561	17:32:25.894083	172.16.1.35	66.39.22.157	TCP	54	3537 → 61092 [ACK] Seq=1 Ack=5809 Win=65535 Len=0
1562	17:32:25.899829	66.39.22.157	172.16.1.35	FTP-DATA	794	FTP Data: 740 bytes (PASV) (LIST)
1563	17:32:25.899969	172.16.1.35	66.39.22.157	TCP	54	3537 → 61092 [ACK] Seq=1 Ack=6550 Win=64795 Len=0
1564	17:32:25.900093	172.16.1.35	66.39.22.157	TCP	54	3537 → 61092 [FIN, ACK] Seq=1 Ack=6550 Win=64795 Len=0
1565	17:32:25.932910	172.16.1.35	66.39.22.157	TCP	54	3534 → 21 [ACK] Seq=112 Ack=726 Win=64810 Len=0

7.

- 66.39.22.157 [linux-wlan.org] [ftp.linux-wlan.org] (FreeBSD)
  - IP: 66.39.22.157
  - MAC: 000D72351E11
  - NIC Vendor: 2Wire Inc
  - MAC Age: 6/1/2003
  - Hostname: linux-wlan.org, ftp.linux-wlan.org
  - OS: FreeBSD
  - TTL: 47 (distance: 17)
  - Open TCP Ports: 21 (FtpControl) 61090 61092 61095 61096
  - Sent: 104 packets (33,380 Bytes), 0.00% cleartext (0 of 0 Bytes)
  - Received: 92 packets (4,188 Bytes), 0.00% cleartext (0 of 0 Bytes)
  - Incoming sessions: 8
  - Outgoing sessions: 0
  - Host Details

8.

- 172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS<00>] (Windows)
  - IP: 172.16.1.35
  - MAC: 0040CA7019A3
  - NIC Vendor: FIRST INTERNAT'L COMPUTER, INC
  - MAC Age: 9/8/2000
  - Hostname: KaufmanUpstairs, KAUFMANUPSTAIRS<00>
  - OS: Windows
    - Ettercap: Windows 2000 Professional SP4 (100.00%)
    - p0f (NetSA): Windows 2000 SP4, XP SP1+, 2003 [Windows] (100.00%)
    - Satori DHCP: Windows - Windows XP (16.67%) Xerox WorkCentre - 7545 [Printer] [Xerox Corp.] (8.33%) Windows - Windows Vista (8.33%) Windows - Windows Server 2012 R2 (8.33%) Win
    - Satori TCP: Windows - Windows 7 (33.33%) Windows - Windows XP (16.67%) Windows - Windows Vista (16.67%) Windows - Windows 2000 (16.67%) Windows - Windows 10 (16.67%)
  - TTL: 128 (distance: 0)
  - Open TCP Ports:
  - Sent: 1125 packets (113,766 Bytes), 0.00% cleartext (0 of 0 Bytes)
  - Received: 1176 packets (650,621 Bytes), 0.00% cleartext (0 of 0 Bytes)
  - Incoming sessions: 0
  - Outgoing sessions: 97
  - Host Details

9.

1451	17:32:20.886743	2Wire_35:1e:11	Broadcast	ARP	60	Who has 172.16.1.37? Tell 172.16.0.1
1452	17:32:20.906413	2Wire_35:1e:11	Broadcast	ARP	60	Who has 172.16.1.39? Tell 172.16.0.1
1453	17:32:20.926121	2Wire_35:1e:11	Broadcast	ARP	60	Who has 172.16.1.34? Tell 172.16.0.1
1454	17:32:20.956031	2Wire_35:1e:11	Broadcast	ARP	60	Who has 172.16.1.35? Tell 172.16.0.1
1455	17:32:20.956048	FirstInt_70:…	2Wire_35:1e:11	ARP	42	172.16.1.35 is at 00:40:ca:70:19:a3

10.

Wireshark · Resolved Addresses

Hosts | Ports | Capture File Comments

aol | All entries

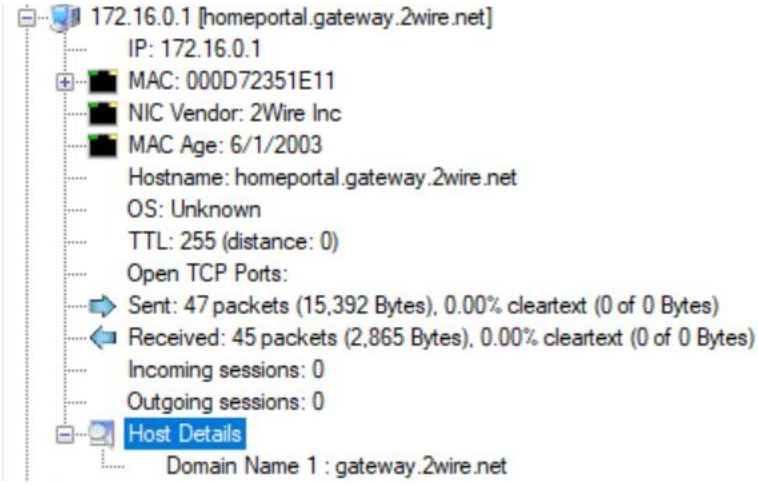
Address	Name
152.163.11.14	2.internet.aol.com
152.163.11.78	2.internet.aol.com
205.188.78.137	2.internet.aol.com
205.188.65.205	2.internet.aol.com
205.188.68.13	2.internet.aol.com
152.163.0.26	aol-23a.aol.com
205.188.64.153	aol-23b.aol.com
152.163.159.232	dns-01.ns.aol.com
205.188.157.232	dns-02.ns.aol.com
152.163.5.135	ipt-rtce07.dial.aol.com

Close

11.

172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS<00>] (Windows)
IP: 172.16.1.35
MAC: 0040CA7019A3
NIC Vendor: FIRST INTERNAT'L COMPUTER, INC
MAC Age: 9/8/2000
Hostname: KaufmanUpstairs, KAUFMANUPSTAIRS<00>
OS: Windows
Ettercap: Windows 2000 Professional SP4 (100.00%)
p0f (NetSA): Windows 2000 SP4, XP SP1+, 2003 [Windows] (100.00%)
Satori DHCP: Windows - Windows XP (16.67%) Xerox WorkCentre - 7545 [Printer] [Xerox Corp.] (8.33%) Windows - Windows Vista (8.33%) Windows - Windows Server 2012 R2 (8.33%) Windc
Satori TCP: Windows - Windows 7 (33.33%) Windows - Windows XP (16.67%) Windows - Windows Vista (16.67%) Windows - Windows 2000 (16.67%) Windows - Windows 10 (16.67%)
TTL: 128 (distance: 0)
Open TCP Ports:
Sent: 1125 packets (113,766 Bytes), 0.00% cleartext (0 of 0 Bytes)
Received: 1176 packets (650,621 Bytes), 0.00% cleartext (0 of 0 Bytes)

12.


172.16.0.1 [homeportal.gateway.2wire.net]
IP: 172.16.0.1
MAC: 000D72351E11
NIC Vendor: 2Wire Inc
MAC Age: 6/1/2003
Hostname: homeportal.gateway.2wire.net
OS: Unknown
TTL: 255 (distance: 0)
Open TCP Ports:
Sent: 47 packets (15,392 Bytes), 0.00% cleartext (0 of 0 Bytes)
Received: 45 packets (2,865 Bytes), 0.00% cleartext (0 of 0 Bytes)
Incoming sessions: 0
Outgoing sessions: 0
Host Details
Domain Name 1 : gateway.2wire.net

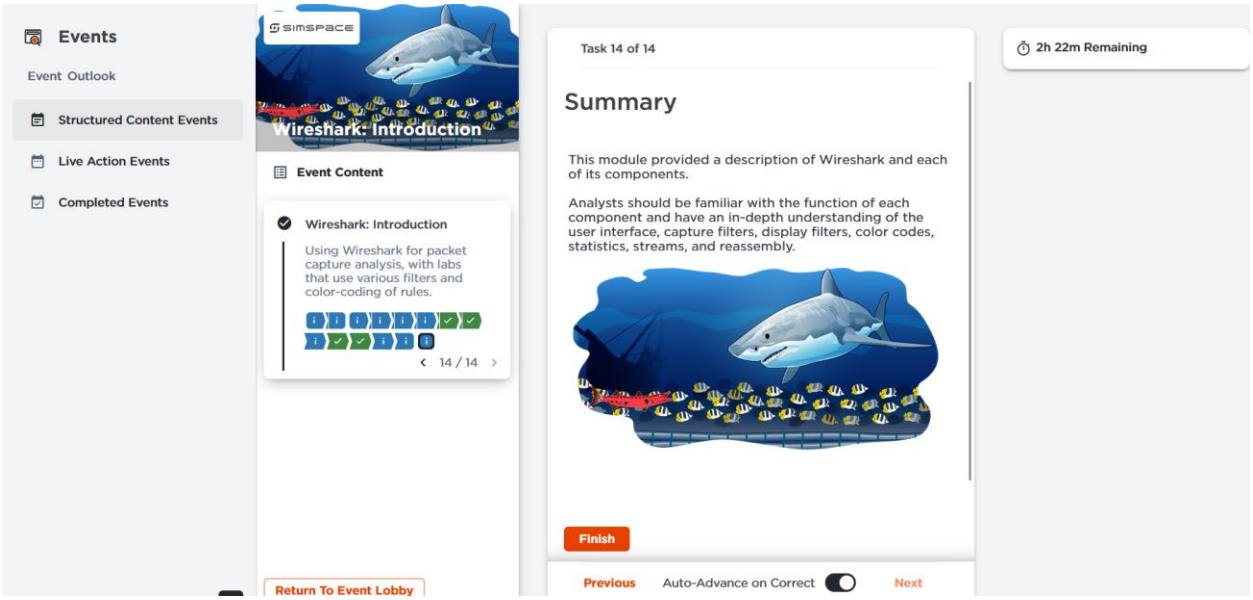
13.

172.16.1.37 [DVR-8525.local]
- IP: 172.16.1.37
- MAC: 00062526D622
- NIC Vendor: The Linksys Group, Inc.
- MAC Age: 6/4/2001
- Hostname: DVR-8525.local
- OS: Unknown
- TTL: 64 (distance: 0)
- Open TCP Ports:
- Sent: 14 packets (2,976 Bytes), 0.00% cleartext (0 of 0 Bytes)
- Received: 0 packets (0 Bytes), 0.00% cleartext (0 of 0 Bytes)
- Incoming sessions: 0
- Outgoing sessions: 0
- Host Details
  - Queried DNS names : Now Playing on DVR 8525._tivo-videos._tcp.local

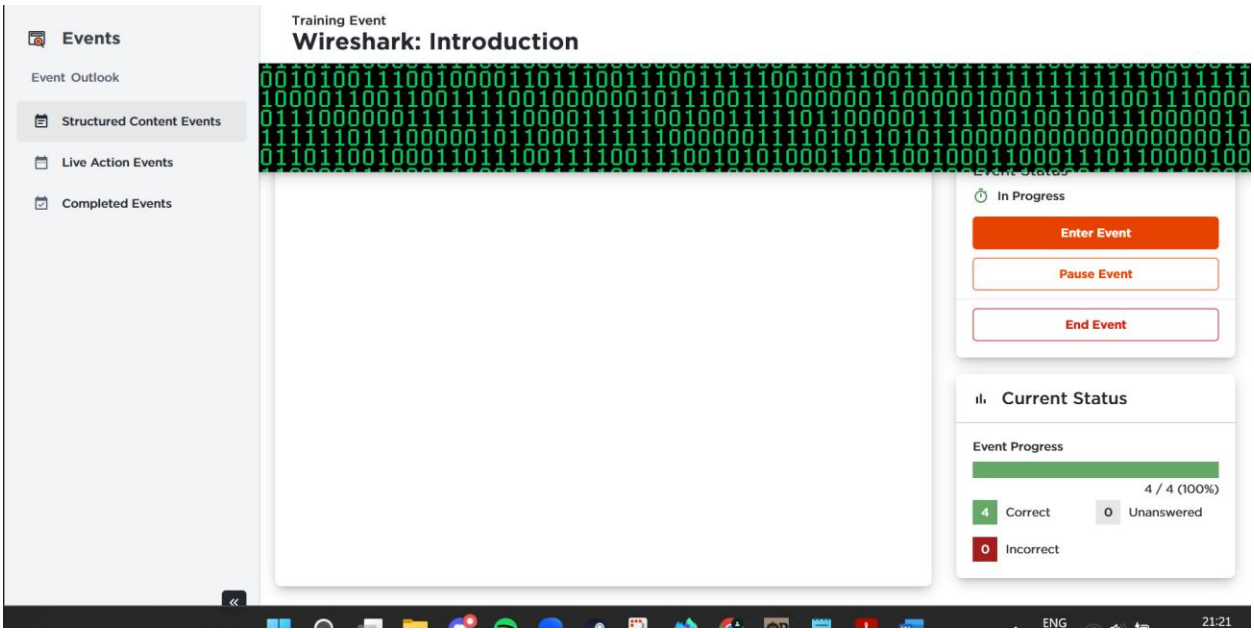
14.

172.16.1.39 [KaufmanUpstairs] [KAUFMANUPSTAIRS<00>] [KAUFMANUPSTAIRS] (Windows)
- IP: 172.16.1.39
- MAC: 0040CA7019A3
- NIC Vendor: FIRST INTERNAT'L COMPUTER, INC
- MAC Age: 9/8/2000
- Hostname: KaufmanUpstairs, KAUFMANUPSTAIRS<00>, KAUFMANUPSTAIRS
- OS: Windows
- TTL: 128 (distance: 0)
- Open TCP Ports:
- Sent: 10 packets (1,570 Bytes), 0.00% cleartext (0 of 0 Bytes)
- Received: 0 packets (0 Bytes), 0.00% cleartext (0 of 0 Bytes)
- Incoming sessions: 0
- Outgoing sessions: 0
- Host Details
  - Queried NetBIOS names : WPAD<00>,MSHOME<1B>
  - Domain Name 1 : MSHOME
  - DHCP Vendor Code 1 : MSFT 5.0
  - Device Family : Axis Communications
  - Device Category : IP Network Camera

15.


Events
Event Outlook
Structured Content Events
Live Action Events
Completed Events
simspace
Wireshark: Introduction
Event Content
Wireshark: Introduction
Using Wireshark for packet capture analysis, with labs that use various filters and color-coding of rules.
14 / 14
Return To Event Lobby
Task 14 of 14
Summary
This module provided a description of Wireshark and each of its components.
Analysts should be familiar with the function of each component and have an in-depth understanding of the user interface, capture filters, display filters, color codes, statistics, streams, and reassembly.
Finish
Previous
Auto-Advance on Correct
Next
2h 22m Remaining

16.


Events
Event Outlook
Structured Content Events
Live Action Events
Completed Events
Training Event
Wireshark: Introduction
In Progress
Enter Event
Pause Event
End Event
Current Status
Event Progress
4 / 4 (100%)
4 Correct
0 Unanswered
0 Incorrect
ENG
21:21

Tools Used

- Wireshark
- NetworkMiner
- Firefox Search Engine